
Power Consumption for Quadcopter Drone

ISYE 6414 – Regression Analysis

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1. Introduction

This project analyzes power consumption in quadcopter drone performance across different flight conditions. Unmanned Aerial Vehicles (UAVs), commonly known as drones, are widely used across multiple sectors, including agriculture for crop monitoring and spraying, search and rescue operations in difficult terrains, environmental monitoring such as tracking wildlife and pollution, healthcare delivery by transporting medical supplies to remote areas, and surveillance for security and monitoring purposes. These are a few examples, with many more domains continuing to adopt this technology.

Analysis of power is a key factor in drone flight because it directly dictates the flight time, payload capacity, performance, operational safety, and cost efficiency. Since drones are generally powered by limited-capacity onboard batteries (like Lithium Polymer), understanding and managing power consumption is essential to prevent mid-flight failures and optimize performance.

2. Problem Statement

The objective of this project is to develop and validate a statistical model that quantifies the relationship between flight conditions and power consumption in a quadcopter drone. Power consumption, defined as $P=V \cdot I$, serves as the response variable and is a key indicator of energy efficiency, flight endurance, and system performance. The analysis utilizes high-frequency telemetry data collected from multiple drone flights. A multiple linear regression framework is employed to model power consumption as a function of relevant predictor variables, including altitude, velocity, acceleration, wind conditions, and flight phase. In addition, individual t-tests are used to evaluate the significance of each predictor variable. To assess the statistical significance of the model, an overall hypothesis test is conducted, where the null hypothesis states that none of the predictors influence power consumption, against the alternative hypothesis that at least one coefficient is nonzero:

Null Hypothesis (H_0): $\beta_1 = \beta_2 = \dots \beta_k = 0$; *no predictors are statistically significant.*

Alternative Hypothesis (H_a): *at least one predictor variable has statistically significant effect on power consumption*

The study emphasizes rigorous variable selection techniques (e.g., subset selection, stepwise regression, or regularization methods) to identify significant predictors while mitigating issues such as multicollinearity, overfitting, and model instability.

Given the time-dependent and high-resolution nature of the dataset, appropriate preprocessing steps—such as down-sampling, noise reduction, and residual diagnostics—are implemented to ensure compliance with regression assumptions, including linearity, independence, homoscedasticity, and normality of errors.

Measurement of statistical performance were performed by observing values such as R^2 , adjusted R^2 , AIC, BIC, and validation techniques. The resulting model is intended to provide accurate predictions of power consumption and offer insights that can support optimization of flight operations, energy management, and control system design.

3. Data Description

The original dataset was collected and published by Carnegie Mellon University and consists of more than 250,000 time-stamped sensor readings for 209 flights. The measurements captured are battery voltage, battery voltage, GPS positions (x, y, z), axis orientation (x, y, z), axis velocity (x, y, z), axis velocity (x, y, z), axis angular velocity (x, y, z), axis linear acceleration (x, y, z), wind speed, wind angle, speed and payload mass. Deduced from the original dataset, power consumption (“Power”) became the response variable for this analysis.

Table 1 – Variables with Definitions			
Variables	Variable Type	Data Type	Units
Power	Dependent	Quantitative - Continuous	Watts
Time	Independent	Quantitative - Continuous	seconds
Altitude	Independent	Quantitative - Continuous	m
Speed	Independent	Quantitative - Continuous	m/s
Payload Mass	Independent	Quantitative - Categorical	lbs.
Position	Independent	Quantitative - Continuous	m
Velocity	Independent	Quantitative - Continuous	m/s
Acceleration	Independent	Quantitative - Continuous	lbs.
Orientation	Independent	Quantitative - Continuous	Degrees
Wind speed	Independent	Quantitative - Continuous	m/s
Wind angle	Independent	Quantitative - Continuous	Degrees

Figure 1. Table of Data Types

The raw data was recorded at sub-second intervals, producing severe within-flight autocorrelation—consecutive readings within a single flight are nearly identical. Two independent approaches were used to address this autocorrelation, Stepwise and LASSO.

Approach 1 – Stepwise

Three representative observations were selected from each flight corresponding to takeoff (phase 1), cruise (phase 2), and landing (phase 3), yielding 628 observations. This introduced a new predictor variable Flight Phase, which is a qualitative variable representing the flight dynamics at different physical and environmental conditions. These flight phase categorical variables are important because they explain the differences in power consumption between each phase.

Approach 2 – LASSO

Each variable was averaged across all time steps within each flight, collapsing the 250,000 rows to 209 independent flight-level observations. This step eliminates within-flight autocorrelation entirely and treats each flight as a single statistical unit.

The identifiers (flight ID), time stamps, and GPS coordinates were removed, leading to computation of power (voltage * current). Wind speed, wind angle, orientation w, velocity (x, y, z), linear acceleration (x, y, z), speed, and payload were included in the predictors.

Furthermore, in the Stepwise regression the ‘time’ was captured in milliseconds. To counter multicollinearity and potential inaccurate results, a systematic-random approach was made. Sparsing the data set into three different flight phases: take off, cruise, and landing. Random data points in those phases of the flights were used in the clean dataset. Each flight was now comprised of 3 data points, allowing the signal-to-noise ratio to be consistent. Whereas, the LASSO regression was performed by taking the average of each flight, totaling to 209 observations.

The descriptive statistics captured a negative power consumption that hit a minimum at -4.36 watts, which is counter intuitive. Corresponding to grounded states captured in the data, those data points were removed from the dataset. Additionally, detected outliers caused flights below 200 Watts to be removed for the LASSO dataset.

3.1 Exploratory Analysis

Approach 1 – Stepwise

Prior to fitting the model, analysis was used to evaluate relationships between the independent variables and the dependent variable, total power. A correlation matrix was generated using pairwise observations. The correlation plot was used to determine the strength of the linear relationships, identify possible multicollinearity, and determine if multiple predictors were capturing similar behavior of the drone.

This analysis helped to show the physically meaningful variable related to payload, vertical motion, orientation, and flight phase. The correlation matrix provided a way to look for multicollinearity between the orientation and velocity predictors. These findings were used in determining the model diagnostics and variable selection process.

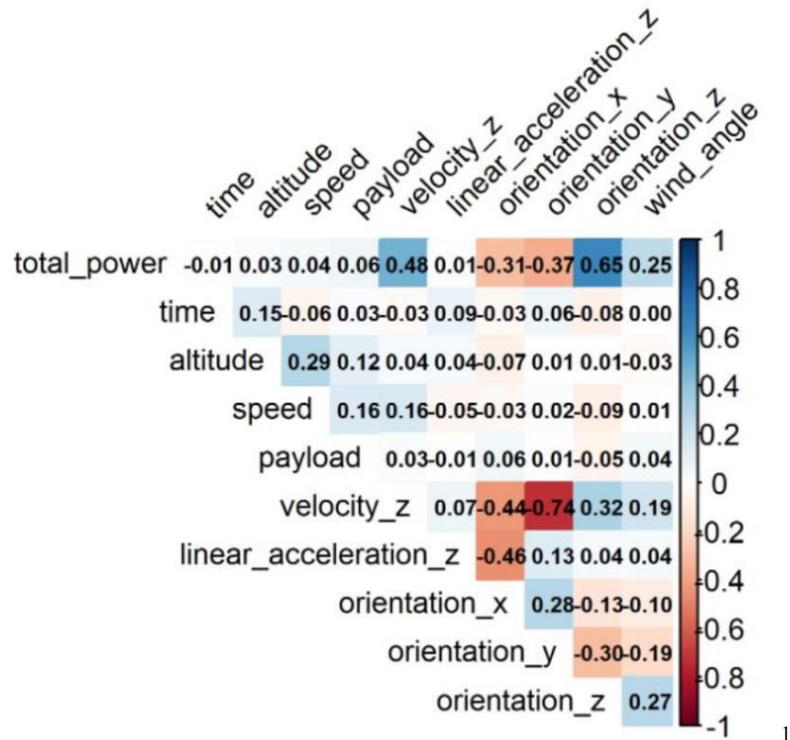


Figure 2. Correlation Matrix Output from R (Stepwise)

Approach 2 – LASSO

Descriptive statistics and visual tools were used to understand data and relationships. The dependent variables. Prior to model fitting, a formal power analysis was conducted using the pwr package in R. With Cohen's $f^2 = 0.15$ (medium effect size), 10 candidate predictors, and $\alpha = 0.05$, the analysis yielded a statistical power of 0.983 for the available 209 flights—well above the conventional 0.80 threshold. The minimum sample size required for 80% power was 118 observations. This confirmed that the aggregated dataset was sufficiently large to detect meaningful predictor effects. A histogram of power consumption (after outlier removal of power

<200) showed a roughly normal distribution centered between 350–450 watts with a slight left skew.

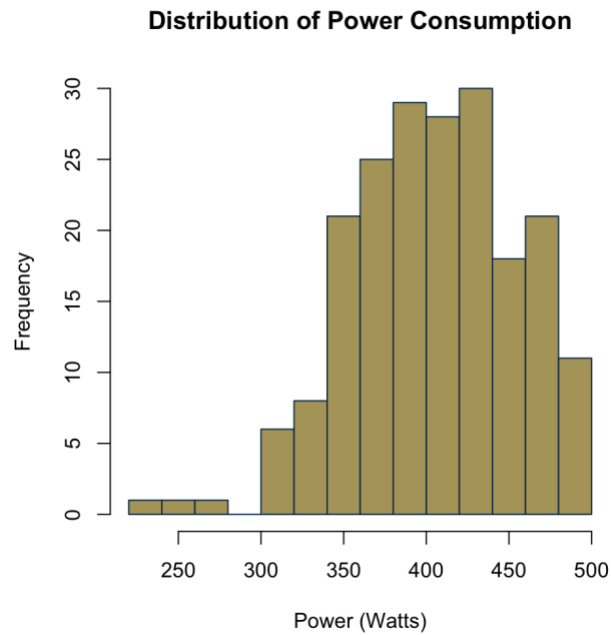


Figure 3. Histogram of Power Consumption (LASSO)

A correlation matrix was computed across all numeric predictors. The variables most strongly correlated with power consumption were: payload (+0.58), wind angle (+0.23), linear acceleration (y) (−0.18), orientation (y) (−0.15), speed (+0.14), wind speed (+0.11), and velocity (z) (−0.10). The correlation matrix also revealed high inter-predictor correlations between orientation (z) and orientation (w) (≈ 0.95) and between velocity (y) and velocity (z) (≈ 0.80), flagging potential multicollinearity.

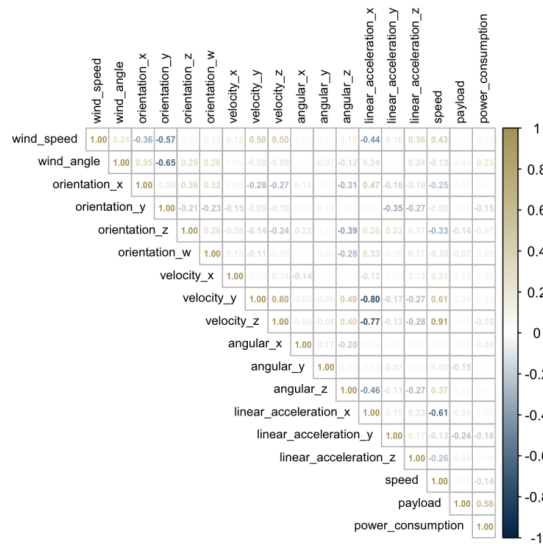


Figure 3. Correlation Matrix (LASSO)

Boxplots of power consumption versus payload (0, 250, 500, 750 grams) confirmed a clear monotonic increase in median power with heavier payloads, consistent with the correlation results. The 750g category contained only one flight and was noted as unreliable. Scatterplots with fitted trend lines were generated for power versus wind speed (positive slope), wind angle (positive slope), and linear acceleration (negative slope). These scatterplot visuals confirmed the results in the correlation matrix, properly identifying accurate predictors.

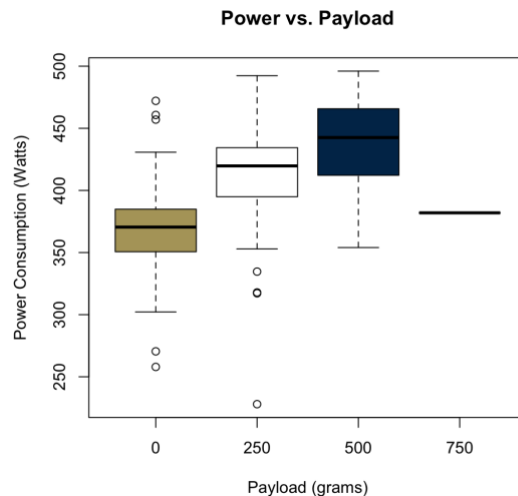


Figure 4. Boxplots of Power vs Payload (LASSO)

4. Analysis

Approach 1 – Stepwise

A full model was fit on the 80/20 training split ($n \approx 502$) using 11 predictors plus the flight phase factor. 80% of the data was used to fit and select the model, while 20% was held to test and validate. Using the Train-Test split helped to determine if the selected model performs well on any unseen observations to help improve the RSME of the final mode. The initial full model included all predictor variables allowing analysis of the combined relevant predictors before reducing the model. This model achieved $R^2 = 0.85$, adjusted $R^2 = 0.85$, and $F = 231$ ($p < 2.2e-16$). The high R^2 is partly attributable to the flight phase variable, which captures the large systematic difference between takeoff/cruise and landing power states.

```
Call:
lm(formula = total_power ~ time + altitude + speed + payload +
    velocity_z + linear_acceleration_z + orientation_x + orientation_y +
    orientation_z + wind_angle + flight_phase, data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-426.50  -23.66   -6.46   11.93   644.40

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -569.93424    141.40760   -4.030 6.47e-05 ***
time              0.04037     0.17026    0.237 0.812667
altitude         0.02945     0.16360    0.180 0.857206
speed            0.37748     1.43122    0.264 0.792085
payload          0.06283     0.02008    3.129 0.001859 **
velocity_z       31.28283     9.40328    3.327 0.000945 ***
linear_acceleration_z -57.58281    14.40415   -3.998 7.40e-05 ***
orientation_x    -389.68966    174.10035   -2.238 0.025656 *
orientation_y     279.55776    125.60720    2.226 0.026498 *
orientation_z     23.13748     7.26534    3.185 0.001543 **
wind_angle        0.04790     0.05478    0.874 0.382320
flight_phase2     416.52073    21.22223   19.627 < 2e-16 ***
flight_phase3     -8.90241     33.48109   -0.266 0.790434
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 93.13 on 483 degrees of freedom
Multiple R-squared:  0.8516,    Adjusted R-squared:  0.8479
F-statistic: 231 on 12 and 483 DF, p-value: < 2.2e-16
```

Figure 5. Initial Trained Full Model output in R (Stepwise)

A step wise regression was implemented using a bidirectional selection approach. This allowed the predictor variables to enter and exit the model iteratively based on their statistical significance and model fit criteria. The implementation started with the full model containing all predictors including the categorical flight phase variable. For each iteration, variables were selected or removed using hypothesis testing on coefficients or t-tests and changes in the overall model performance. This method is used to reduce the full model to a more parsimonious model and ensures the model retains explanatory power.

The full model was adjusted to 8 predictors: payload, velocity (z), linear acceleration (z), orientation (x), orientation (y), orientation (z), and flight phase. The final stepwise model that retained the most significant predictors and eliminated the least useful variables was:

$$\begin{aligned} \text{Total Power} = & -559.75 + 0.065(\text{payload}) + 32.31(\text{velocity_z}) - \\ & 57.78(\text{linear_acceleration_z}) - 383.07(\text{orientation_x}) + 284.53(\text{orientation_y}) + \\ & 23.83(\text{orientation_z}) + 420.93(\text{flight_phase2}) - 1.49(\text{flight_phase3}) \end{aligned}$$

```
Call:
lm(formula = total_power ~ payload + velocity_z + linear_acceleration_z
+ orientation_x + orientation_y + orientation_z + flight_phase,
data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-432.51  -23.87   -6.73   11.88   648.48

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -559.74857   139.69487  -4.007 7.11e-05 ***
payload         0.06542    0.01968    3.324 0.000954 ***
velocity_z     32.31301    9.04530    3.572 0.000389 ***
linear_acceleration_z -57.77913   14.21336  -4.065 5.60e-05 ***
orientation_x  -383.07474   172.54749  -2.220 0.026871 *
orientation_y   284.53190   121.71405    2.338 0.019806 *
orientation_z   23.82536    7.11858    3.347 0.000880 ***
flight_phase2  420.93503   13.59464   30.963 < 2e-16 ***
flight_phase3   -1.49111   10.27486   -0.145 0.884674

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92.84 on 487 degrees of freedom
Multiple R-squared:  0.8513,    Adjusted R-squared:  0.8489
F-statistic: 348.5 on 8 and 487 DF,  p-value: < 2.2e-16
```

Figure 6. Final Stepwise Model Output from R

The final stepwise model shows that the most important factors in power consumption are payload and vertical motion. The positive coefficient on payload makes sense because heavier payloads need additional thrust and that increases the demand for power. The positive coefficient on velocity z shows that vertical upward movement results in higher power consumption. The dynamics of the system are accounted for in the acceleration and orientation variables. Having a large coefficient for flight_phase2 indicates the flight condition for cruise is significantly impacting the power consumption. This is due to the fact that when the drone is at cruising altitude it is not moving in the z direction.

Approach 2 – LASSO

LASSO (Least Absolute Shrinkage and Selection Operator) regression was applied with 10-fold cross-validation to determine the optimal regularization parameter λ . At $\lambda_{\min} = 0.907$, LASSO retained 14 of 17 predictors, setting the coefficients of orientation (y), velocity (x), velocity (z), and linear acceleration (x) to exactly zero. The cross-validation plot confirmed that mean squared error remained stable across a wide range of λ values, indicating that a small subset of variables carried most of the predictive power.

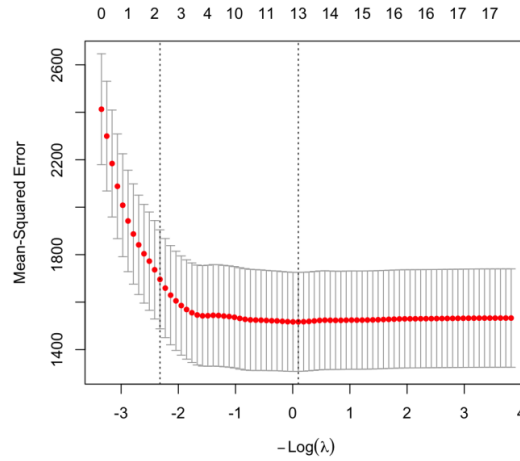


Figure 7. LASSO regression output

In the end, 14 variables were retained by LASSO in the multi linear regression fit. The model achieved $R^2 = 0.51$, adjusted $R^2 = 0.48$, and an overall F-statistic p-value $< 2.2e-16$, confirming that the model was statistically significant. However, only four predictors achieved individual significance at $\alpha = 0.05$: payload ($p < 2e-16$), wind angle ($p = 0.002$), wind speed ($p = 0.042$), and velocity (z) ($p = 0.016$). The reduced model performed better than the full model. The reduced model fits using only the four statistically significant predictors: wind speed, wind angle, velocity (z), and payload. All four predictors remained significant (all $p < 0.04$), with an adjusted $R^2 = 0.41$ and $F = 35.81$ ($p < 2.2e-16$). An ANOVA comparison between the full 14-predictor model and the reduced 4-predictor model yielded $p = 0.0004$, indicating that the full model did explain significantly more variance. However, the reduced model was selected as the final LASSO model and the adjusted R^2 loss was modest (0.48 to 0.41). The final LASSO model was:

$$\text{Power Consumption} = 307.63 + 8.20(\text{wind speed}) + 0.25(\text{wind angle}) - 50.17(\text{velocity (z)}) + 0.14(\text{payload})$$

```

Call:
lm(formula = power_consumption ~ wind_speed + wind_angle + velocity_z +
    payload, data = avgflight_data_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-149.590  -18.515    1.825   23.731   92.775

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 307.63070    16.54024   18.599 < 2e-16 ***
wind_speed    8.20309     3.63864    2.254 0.025281 *
wind_angle    0.25033     0.07224    3.465 0.000652 ***
velocity_z   -50.16771    18.47688   -2.715 0.007219 **
payload       0.13789     0.01268   10.876 < 2e-16 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 37.59 on 195 degrees of freedom
Multiple R-squared:  0.4235,    Adjusted R-squared:  0.4117
F-statistic: 35.81 on 4 and 195 DF,  p-value: < 2.2e-16

```

Figure 8. Reduced Model (LASSO)

5. Diagnostic Tests

Approach 1 – Stepwise

The final model was compared to the full model using Analysis of Variance (ANOVA). Comparing the models shows if the variables removed by stepwise selection contributed to any statistically meaningful explanatory power. The p value was tested for significance, and it was concluded that the reduced model should be selected since it has similar explanatory power using fewer predictors. After analyzing the VIF values, they were all less than 2.65, which is less than the critical threshold of 5, leading to conclude that there were no multicollinearity issues. Another tool used to validate was the Durbin-Watson test, with a value of 1.91 and p value of .16, failing to reject the null hypothesis of no autocorrelation. The four standard diagnostic plots (Q-Q plot, residuals vs fitted, scale-location, residuals vs leverage) were plotted and checked for unusual observations. The Q-Q plot showed mild deviations but not severe enough to invalidate the model. The scale location plot showed some instability in residuals suggesting possible heteroscedasticity. The residuals vs leverage plot showed observations 452, 455, and 208 flagged as high-influence points, but none exceeding Cook's distance thresholds. The diagnostic tests showed some instability, so the stepwise model should be used as an explanatory model and not the model with the strongest predictive power.

Approach 2 – LASSO

The Shapiro-Wilk test on the reduced model residuals yielded $W = 0.973$, $p = 0.0007$. While statistically significant, the W-statistic is very close to 1.0, indicating only mild departure from

normality driven by a small number of outlier flights. With $n \approx 200$, the Central Limit Theorem ensures robust inference despite this result. The Breusch-Pagan test yielded $BP = 8.96$, $p = 0.062$ for the reduced model—borderline but not significant at $\alpha = 0.05$, indicating approximately constant variance. For the full 14-predictor model, $BP = 30.50$, $p = 0.007$, indicating heteroscedasticity. Robust standard errors (HC1) were applied via the sandwich package. Under robust inference, all four predictors in the reduced model remained statistically significant (payload $p < 2.2e-16$, wind angle $p = 0.002$, velocity (z) $p = 0.007$, wind speed $p = 0.039$). Variance inflation factors for the reduced model were wind speed = 1.48, wind angle = 1.13, velocity (z) = 1.41, payload = 1.00. All values are well below the conventional threshold of 5, confirming no multicollinearity concerns. The LASSO reduced model showed generally well-behaved residuals with a few outlier flights (observations 85, 87, 134). The visuals can be found in the appendix section.

6. Validation

Approach 1 – Stepwise

The stepwise model was retrained on 80% of the phase-sampled data (`set.seed(42)`) and tested on the held-out 20%. The out-of-sample RMSE was 91.51 watts. The higher RMSE is attributable to the inclusion of takeoff and landing phases, which introduce greater variance in power readings. The Actual vs. Predicted plot showed good clustering at the high-power (400–600W cruise) end, but systematic prediction errors for the low-power (0–50W grounded/landing) observations. The validation concludes that this model is only good for explanatory purposes since it captures the power trends due to the flight phase but has weak predictive power during the lower power phases and transitional or between flight phases.

Approach 2 – LASSO

The 4-predictor reduced model was retrained on 80% of the aggregated data (`set.seed(123)`) and tested on the remaining 20%. The out-of-sample RMSE was 38.61 watts, representing approximately 10% of the mean power consumption (≈ 388 watts). The Actual vs. Predicted plot showed points clustering around the 45-degree reference line across the 340–460 watt range, confirming reasonable predictive accuracy.

Results	LASSO	Stepwise
Data Reduction	Aggregate flight (n=200)	Phase sampling (n=628)
Selection Method	Predictors are eliminated	Predictors go to 0
Final Predictors	4: wind speed, wind angle, velocity (z), payload	8: payload, velocity (z), linear acceleration (z), orientation (x, y, z), flight phases
Adjusted R ²	0.41	.85
Test RMSE	38.61 W	91.51 W
VIF (max)	4.13	2.63
Heteroscedasticity	Robust SE applied (p = .06)	Not addressed
Autocorrelation	Eliminated aggregation	No issue (p = .16)
Strength	Better prediction, robust	Explains more of the model, higher R ²

Figure 10. Regression technique comparison

7. Conclusions

Two modeling techniques—Stepwise Regression and LASSO—were implemented to ensure both interpretability and predictive robustness. LASSO demonstrated superior predictive performance, achieving a lower RMSE by effectively handling multicollinearity and reducing model complexity. In contrast, Stepwise Regression provided higher interpretability and a better explanation of variability, as reflected by a higher adjusted R².

Both modeling approaches converge on the same core finding: payload and velocity (z) are consistently the most important predictors of drone power consumption. The LASSO approach additionally identified wind speed and wind angle as significant environmental factors. The Stepwise approach captured flight-phase effects and orientation variables that explain a larger share of variance but at the cost of model complexity.

The overall null hypothesis is rejected (F-statistic $p < 2.2e-16$ for both models): at least one predictor significantly influences power consumption. Specifically, the individual hypotheses for payload, velocity (z), wind speed, and wind angle are all rejected at $\alpha = 0.05$.

From an operational perspective, the results suggest that drone operators can most effectively reduce power consumption by minimizing payload mass where possible and maintaining steady, moderate vertical velocities to avoid energy-intensive altitude changes. Planning flight paths to minimize headwind exposure (wind speed and unfavorable wind angles) offers additional efficiency gains.

The LASSO model's adjusted R^2 of 0.41 indicates that approximately 59% of the variance in power consumption remains unexplained, likely due to unobserved factors such as battery degradation, temperature, and motor efficiency. The signal to-noise for LASSO ignores a lot of the noise because it is only the aggregate of each flight. The aggregation approach also masks within-flight dynamics. The Stepwise model's higher R^2 (0.85) is partly driven by the flight phase variable, which captures a structural feature of the data rather than a controllable flight parameter. The results consistently identified payload and vertical velocity (Z-axis) as the most influential factors driving power consumption. Payload directly increases the total weight of the drone, requiring greater thrust from the motors to maintain lift, which leads to higher power consumption. Similarly, increases in vertical velocity—particularly during ascent—require additional power to overcome gravitational forces and generate upward acceleration, significantly increasing battery usage. These physical relationships explain why both variables emerged as dominant predictors across modeling approaches.

From a practical perspective, reducing payload and avoiding aggressive vertical movements can significantly decrease energy consumption and extend flight time. Maintaining smooth and stable flight conditions further minimizes unnecessary power usage and optimizes flight performance.

Future analyses could incorporate nonlinear terms or interaction effects (e.g., payload $\times$ wind speed), apply Ridge Regression or Elastic Net for comparison, explore Random Forest or gradient boosting for non-linear relationships, or use mixed-effects models on the raw time-series data to capture both within-flight and between-flight variance simultaneously. A re-work of the LASSO regression with a systematic dataset (like the one used for Stepwise) could improve the adjusted R^2 , explaining more of the data.

This project developed a statistical framework to analyze and predict drone power consumption under varying flight conditions using a Multiple Linear Regression (MLR) approach. Leveraging the Carnegie Mellon drone flight dataset, the study examined how operational and environmental variables influence energy usage, with the goal of improving flight efficiency and mission planning.

References

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Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer.

Best Subset, Forward Stepwise or Lasso? Analysis and Recommendations Based on Extensive Comparisons. <https://www.stat.berkeley.edu/~ryantibs/papers/bestsubset-sts.pdf>

Appendix

Appendix A – Exploratory Data Analysis

LASSO (Power Test and Sample Size)

- First test: With our 209 flights and 10 predictors, the power is 0.983 — that means we had a 98.3% chance of detecting a real effect. Anything above 0.80 is considered good.
- Second test: The minimum sample size needed for 80% power with 10 predictors is $v = 107$, which means $n = 107 + 10 + 1 = 118$ flights. We had 209,.
- A power analysis using Cohen's $f^2 = 0.15$ (medium effect size) with 10 predictors at $\alpha = 0.05$ confirmed that our sample of 209 flights provides statistical power of 0.983, well above the conventional threshold of 0.80. The minimum required sample size for 80% power was 118 observations.
- The power test and sample size was used to determine how many observations we needed to make an accurate useful dataset.

LASSO (Descriptive Statistics)

flight	wind_speed	position_x	position_z	orientation_y	orientation_w
Min. : 1.0	Min. : 0.08699	Min. : -79.95	Min. : 0.0	Min. : -0.037775	Min. : 0.2849
1st Qu.: 112.0	1st Qu.: 3.85417	1st Qu.: -79.78	1st Qu.: 285.0	1st Qu.: -0.014789	1st Qu.: 0.4124
Median : 167.0	Median : 4.31650	Median : -79.78	Median : 300.8	Median : -0.009901	Median : 0.6540
Mean : 161.3	Mean : 4.30947	Mean : -76.35	Mean : 288.6	Mean : -0.010554	Mean : 0.5959
3rd Qu.: 219.0	3rd Qu.: 5.05152	3rd Qu.: -79.78	3rd Qu.: 315.3	3rd Qu.: -0.005451	3rd Qu.: 0.6973
Max. : 279.0	Max. : 6.67019	Max. : 0.00	Max. : 334.3	Max. : 0.011385	Max. : 1.0000
time	wind_angle	position_y	orientation_x	orientation_z	velocity_x
Min. : 29.27	Min. : 73.3	Min. : 0.00	Min. : -0.042059	Min. : -0.70583	Min. : -0.73435
1st Qu.: 78.35	1st Qu.: 141.6	1st Qu.: 40.46	1st Qu.: -0.016412	1st Qu.: -0.35188	1st Qu.: -0.18428
Median : 90.84	Median : 168.4	Median : 40.46	Median : -0.010844	Median : -0.17171	Median : -0.08574
Mean : 92.56	Mean : 169.4	Mean : 38.72	Mean : -0.010384	Mean : -0.10923	Mean : -0.11198
3rd Qu.: 102.92	3rd Qu.: 198.1	3rd Qu.: 40.46	3rd Qu.: -0.002605	3rd Qu.: 0.05323	3rd Qu.: -0.02010
Max. : 214.17	..	Max. : 40.46	Max. : 0.015343	Max. : 0.67456	Max. : 0.79887
velocity_z	angular_x	angular_y	angular_z	linear_acceleration_x	linear_acceleration_z
Min. : -0.07686	Min. : -0.0090613	Min. : -7.907e-03	Min. : -0.01653	Min. : -0.7462	Min. : -9.862
1st Qu.: 0.16508	1st Qu.: -0.0028762	1st Qu.: -1.447e-03	1st Qu.: 0.02320	1st Qu.: -0.4379	1st Qu.: -9.811
Median : 0.31819	Median : -0.0013987	Median : 1.191e-04	Median : 0.03145	Median : -0.3434	Median : -9.803
Mean : 0.31314	Mean : -0.0012946	Mean : 9.012e-05	Mean : 0.02937	Mean : -0.3265	Mean : -9.798
3rd Qu.: 0.43890	3rd Qu.: 0.0001224	3rd Qu.: 1.758e-03	3rd Qu.: 0.03845	3rd Qu.: -0.2369	3rd Qu.: -9.791
Max. : 0.83977	Max. : 0.0062495	Max. : 8.740e-03	Max. : 0.05832	Max. : 0.2261	Max. : -9.588
speed	power_consumption			linear_acceleration_y	
Min. : 0.000	Min. : 4.36			Min. : -0.38943	
1st Qu.: 4.000	1st Qu.: 363.34			1st Qu.: -0.06885	
Median : 8.000	Median : 402.38			Median : 0.01567	
Mean : 7.493	Mean : 387.85			Mean : 0.01549	
3rd Qu.: 10.000	3rd Qu.: 439.24			3rd Qu.: 0.08201	
Max. : 12.000	Max. : 496.03			Max. : 0.32631	

LASSO (Regression)

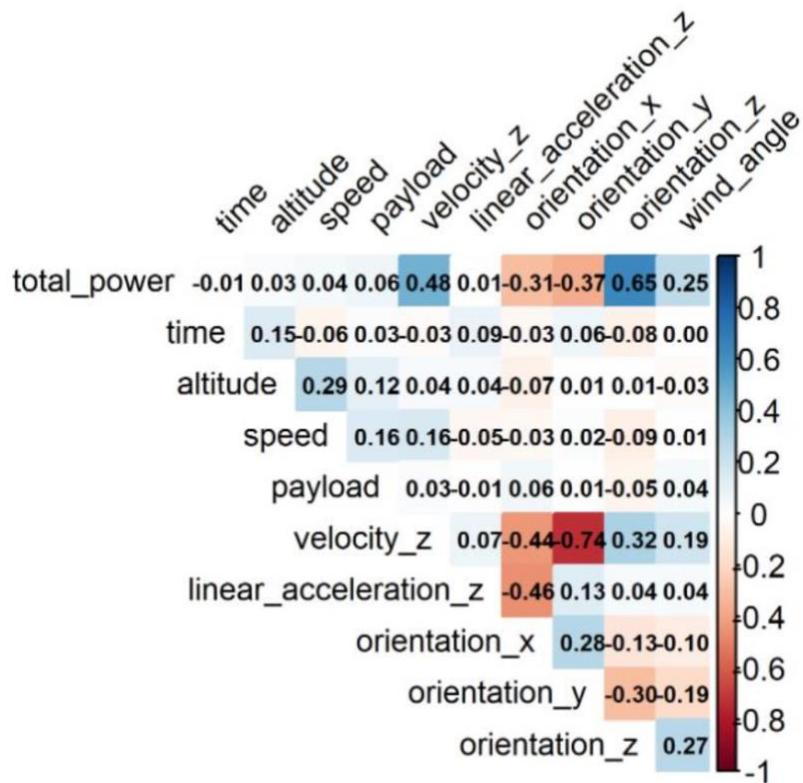
Best lambda: 0.9066487 Lambda 1se: 10.1846

```

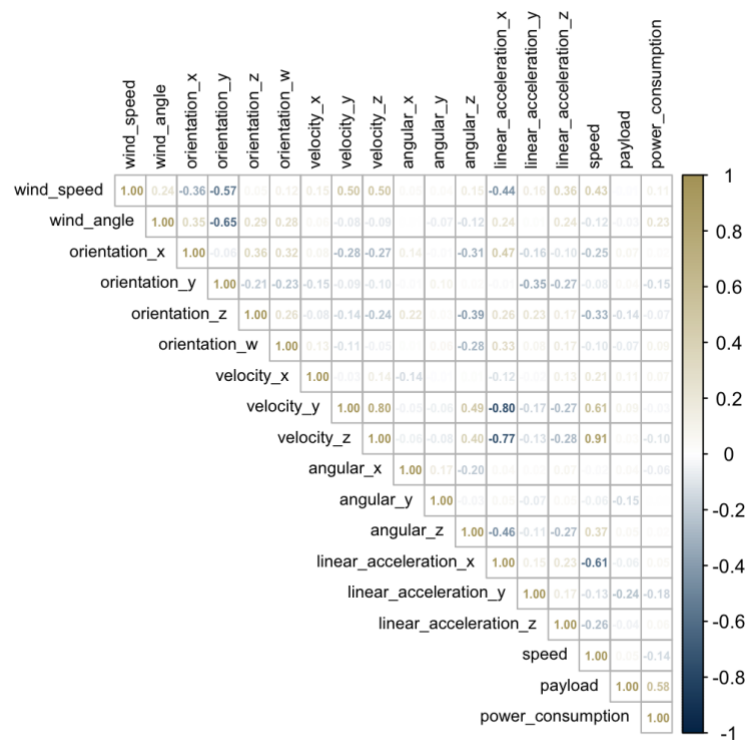
18 x 1 sparse Matrix of class "dgCMatrix"
      lambda.min      coefficient      variable
(Intercept) -1172.0357570 angular_y 1186.0769677 angular_y
wind_speed 9.1836599 angular_x -1181.6606866 angular_x
wind_angle 0.2717590 (Intercept) -1172.0357570 (Intercept)
orientation_x -435.0764526 orientation_x -435.0764526 orientation_x
orientation_y . linear_acceleration_z -150.4666806 linear_acceleration_z
orientation_z -10.2435841 angular_z 45.7823797 angular_z
orientation_w 22.3996878 linear_acceleration_y -29.9666581 linear_acceleration_y
velocity_x . orientation_w 22.3996878 orientation_w
velocity_y -4.1788419 orientation_z -10.2435841 orientation_z
velocity_z . wind_speed 9.1836599 wind_speed
angular_x -1181.6606866 velocity_y -4.1788419 velocity_y
angular_y 1186.0769677 speed -3.7992927 speed
angular_z 45.7823797 wind_angle 0.2717590 wind_angle
linear_acceleration_x . payload 0.1353433 payload
linear_acceleration_y -29.9666581
linear_acceleration_z -150.4666806
speed -3.7992927
payload 0.1353433

```

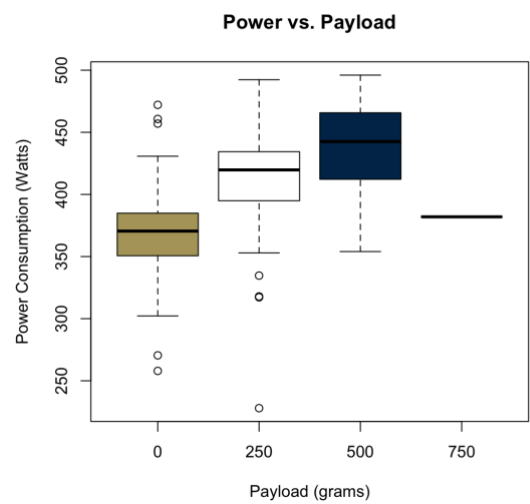
Stepwise (Correlation Matrix)



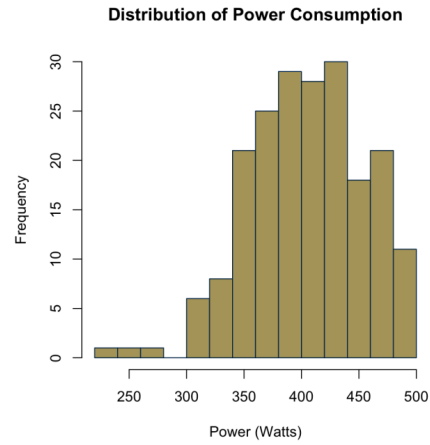
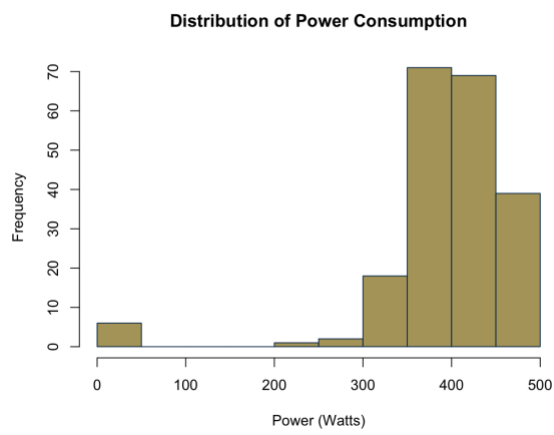
LASSO (Correlation Matrix)



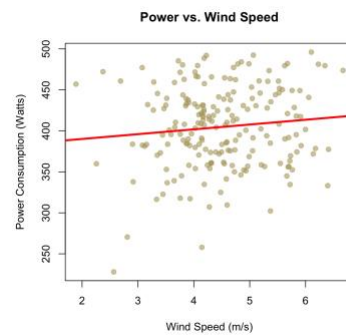
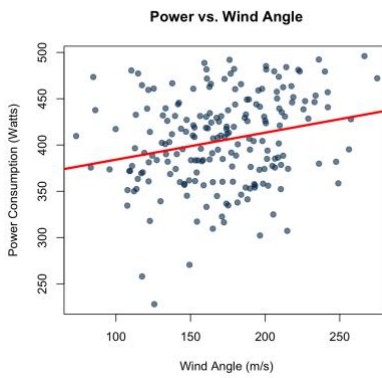
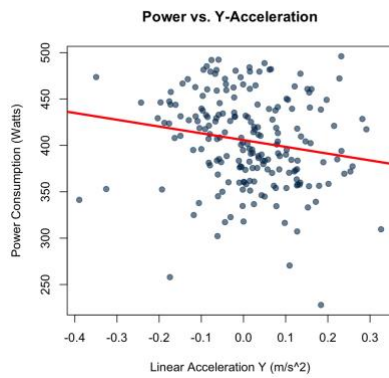
LASSO (Boxplots)



LASSO (Histogram)

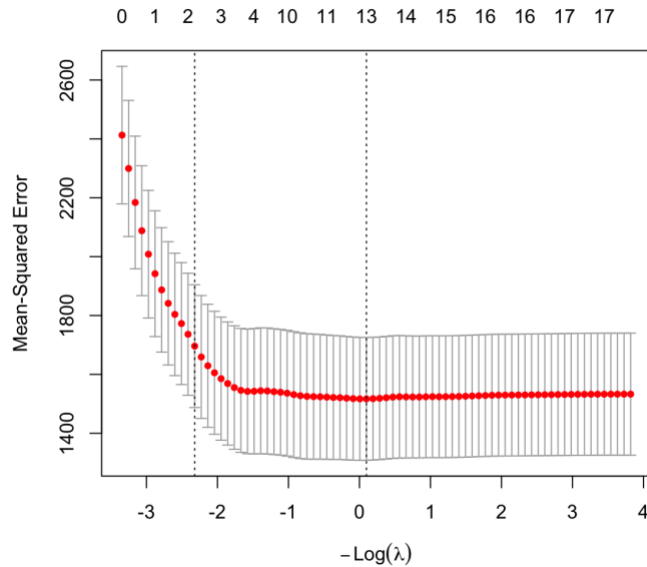


LASSO (Scatterplots)



Appendix B – Modeling

LASSO (Regression)



LASSO (Full Model)

Call:
lm(formula = power_consumption ~ wind_speed + wind_angle + orientation_x +
orientation_z + orientation_w + velocity_y + velocity_z +
angular_x + angular_y + angular_z + linear_acceleration_y +
linear_acceleration_z + speed + payload, data = avgflight_data_clean)

Residuals:
Min 1Q Median 3Q Max
-118.743 -18.545 4.557 23.216 96.770

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.627e+03	1.277e+03	-2.057	0.041091 *
wind_speed	1.506e+01	5.054e+00	2.980	0.003271 **
wind_angle	2.824e-01	8.009e-02	3.526	0.000532 ***
orientation_x	-5.513e+02	3.330e+02	-1.655	0.099530 .
orientation_z	-1.331e+01	1.070e+01	-1.245	0.214880
orientation_w	2.570e+01	1.766e+01	1.455	0.147359
velocity_y	-2.329e+01	9.582e+00	-2.430	0.016039 *
velocity_z	1.018e+02	5.817e+01	1.751	0.081662 .
angular_x	-1.212e+03	1.079e+03	-1.124	0.262493
angular_y	1.511e+03	9.553e+02	1.582	0.115453
angular_z	2.323e+02	2.604e+02	0.892	0.373583
linear_acceleration_y	-4.807e+01	2.471e+01	-1.946	0.053211 .
linear_acceleration_z	-3.001e+02	1.298e+02	-2.311	0.021933 *
speed	-8.589e+00	2.526e+00	-3.400	0.000825 ***
payload	1.431e-01	1.285e-02	11.137	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 35.5 on 185 degrees of freedom
Multiple R-squared: 0.5124, Adjusted R-squared: 0.4754
F-statistic: 13.88 on 14 and 185 DF, p-value: < 2.2e-16

Stepwise (Full Model)

```

===== MODEL SUMMARY =====
> cat("R-squared:", summary(model)$r.squared, "\n")
R-squared: 0.5123524
> cat("Adjusted R-squared:", summary(model)$adj.r.squared, "\n")
Adjusted R-squared: 0.4754494
> cat("F-statistic p-value:", pf(summary(model)$fstatistic[1]
+                               summary(model)$fstatistic[2]
ummary(model)$fstatistic[3],
+                               lower.tail = FALSE), "\n")
F-statistic p-value: 2.952312e-22

Call:
lm(formula = total_power ~ time + altitude + speed + payload +
    velocity_z + linear_acceleration_z + orientation_x + orientation_y +
    orientation_z + wind_angle + flight_phase, data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-426.50  -23.66   -6.46   11.93   644.40

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -569.93424    141.40760   -4.030 6.47e-05 ***
time              0.04037     0.17026    0.237 0.812667
altitude        0.02945     0.16360    0.180 0.857206
speed           0.37748     1.43122    0.264 0.792085
payload         0.06283     0.02008    3.129 0.001859 **
velocity_z      31.28283     9.40328    3.327 0.000945 ***
linear_acceleration_z -57.58281    14.40415   -3.998 7.40e-05 ***
orientation_x   -389.68966    174.10035   -2.238 0.025656 *
orientation_y    279.55776    125.60720    2.226 0.026498 *
orientation_z    23.13748     7.26534    3.185 0.001543 **
wind_angle      0.04790     0.05478    0.874 0.382320
flight_phase2    416.52073    21.22223   19.627 < 2e-16 ***
flight_phase3    -8.90241     33.48109   -0.266 0.790434
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 93.13 on 483 degrees of freedom
Multiple R-squared:  0.8516,    Adjusted R-squared:  0.8479
F-statistic: 231 on 12 and 483 DF, p-value: < 2.2e-16

```

Stepwise (Final Model)

```

Call:
lm(formula = total_power ~ payload + velocity_z + linear_acceleration_z
+
    orientation_x + orientation_y + orientation_z + flight_phase,
    data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-432.51  -23.87   -6.73   11.88   648.48

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -559.74857    139.69487   -4.007 7.11e-05 ***
payload         0.06542     0.01968    3.324 0.000954 ***
velocity_z      32.31301     9.04530    3.572 0.000389 ***
linear_acceleration_z -57.77913    14.21336   -4.065 5.60e-05 ***
orientation_x   -383.07474    172.54749   -2.220 0.026871 *
orientation_y    284.53190    121.71405    2.338 0.019806 *
orientation_z    23.82536     7.11858    3.347 0.000880 ***
flight_phase2    420.93503    13.59464   30.963 < 2e-16 ***
flight_phase3    -1.49111     10.27486   -0.145 0.884674
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92.84 on 487 degrees of freedom
Multiple R-squared:  0.8513,    Adjusted R-squared:  0.8489
F-statistic: 348.5 on 8 and 487 DF, p-value: < 2.2e-16

```

LASSO (Full Model)

studentized Breusch-Pagan test

data: model

BP = 30.502, df = 14, p-value = 0.006505

LASSO (Reduced Model – Final Model)

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	307.630705	19.566371	15.7224	< 2.2e-16
wind_speed	8.203085	3.940693	2.0816	0.038681
wind_angle	0.250329	0.079051	3.1667	0.001790
velocity_z	-50.167709	18.504115	-2.7112	0.007303
payload	0.137892	0.012317	11.1949	< 2.2e-16

(Intercept) ***
wind_speed *
wind_angle **
velocity_z **
payload ***

Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

LASSO (Reduced Model – Final Model)

Call:

```
lm(formula = power_consumption ~ wind_speed + wind_angle + velocity_z +  
    payload, data = avgflight_data_clean)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-149.590	-18.515	1.825	23.731	92.775

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	307.63070	16.54024	18.599	< 2e-16 ***
wind_speed	8.20309	3.63864	2.254	0.025281 *
wind_angle	0.25033	0.07224	3.465	0.000652 ***
velocity_z	-50.16771	18.47688	-2.715	0.007219 **
payload	0.13789	0.01268	10.876	< 2e-16 ***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 37.59 on 195 degrees of freedom

Multiple R-squared: 0.4235, Adjusted R-squared: 0.4117

F-statistic: 35.81 on 4 and 195 DF, p-value: < 2.2e-16

Shapiro-Wilk normality test	studentized Breusch-Pagan test
data: resid(model_reduced)	data: model_reduced
W = 0.97323, p-value = 0.0007208	BP = 8.9567, df = 4, p-value = 0.06219

```

      wind_speed wind_angle velocity_z    payload
VIF: 1.484591  1.129296  1.410319  1.001951

```

Stepwise (VIF)

	GVIF	Df	GVIF^(1/(2*Df))
payload	1.013828	1	1.006890
velocity_z	2.649944	1	1.627865
linear_acceleration_z	1.188115	1	1.090007
orientation_x	1.569862	1	1.252941
orientation_y	2.307913	1	1.519182
orientation_z	1.926491	1	1.387981
flight_phase	2.203136	2	1.218317

LASSO (VIF)

wind_speed	wind_angle	orientation_x
4.058185	2.398543	2.079188
orientation_y	orientation_z	orientation_w
3.301150	1.646234	1.317346
velocity_x	velocity_y	velocity_z
1.255225	4.135905	3.811614
angular_x	angular_y	angular_z
1.176803	1.133247	1.697364
linear_acceleration_y	linear_acceleration_z	payload
1.623847	1.929594	1.168991

LASSO (VIF – Reduced Model – Final Model))

```

      wind_speed wind_angle velocity_z    payload
      1.484591  1.129296  1.410319  1.001951

```

Stepwise (ANOVA)

```

--- ANOVA Model Comparison (Final vs Full) ---
> print(anova(final_model, full_model))
Analysis of Variance Table

Model 1: total_power ~ payload + velocity_z + linear_acceleration_z +
  orientation_x + orientation_y + orientation_z + flight_phase
Model 2: total_power ~ time + altitude + speed + payload + velocity_z +
  linear_acceleration_z + orientation_x + orientation_y + orientation_
z +
  wind_angle + flight_phase
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     487 4197479
2     483 4189083  4      8395.6 0.242 0.9145

```

LASSO (ANOVA)

Analysis of Variance Table

```

Model 1: power_consumption ~ wind_speed + wind_angle + orientation_y +
  velocity_z + linear_acceleration_z + payload
Model 2: power_consumption ~ wind_speed + wind_angle + orientation_x +
  orientation_y + orientation_z + orientation_w + velocity_x +
  velocity_y + velocity_z + angular_x + angular_y + angular_z +
  linear_acceleration_y + linear_acceleration_z + payload
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1      199 544484
2      190 495047  9      49437 2.1082 0.03066 *

```

Analysis of Variance Table

```

Model 1: power_consumption ~ wind_speed + wind_angle + velocity_z + payload
Model 2: power_consumption ~ wind_speed + wind_angle + orientation_x +
  orientation_z + orientation_w + velocity_y + velocity_z +
  angular_x + angular_y + angular_z + linear_acceleration_y +
  linear_acceleration_z + speed + payload
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1      195 275605
2      185 233132 10      42472 3.3704 0.0004635 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

LASSO (Shapiro-Wilk Normality)

Shapiro-Wilk normality test

```

data: resid(model)
W = 0.97735, p-value = 0.002541

```

Appendix C – Diagnostics

Stepwise

```
--- Durbin-Watson (Autocorrelation) ---
> print(dwtest(final_model))

Durbin-Watson test

data: final_model
DW = 1.9126, p-value = 0.1628
alternative hypothesis: true autocorrelation is greater than 0
```

LASSO (RMSE)

Final Model Out-of-Sample RMSE: 38.61 Watts



Cleaned Model RMSE: 38.61 Watts



Stepwise (Full trained model)

```
Call:
lm(formula = total_power ~ time + altitude + speed + payload +
    velocity_z + linear_acceleration_z + orientation_x + orientation_y +
    orientation_z + wind_angle + flight_phase, data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-426.50  -23.66   -6.46   11.93   644.40

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -569.93424    141.40760   -4.030 6.47e-05 ***
time           0.04037     0.17026    0.237 0.812667
altitude       0.02945     0.16360    0.180 0.857206
speed          0.37748     1.43122    0.264 0.792085
payload        0.06283     0.02008    3.129 0.001859 **
velocity_z     31.28283     9.40328    3.327 0.000945 ***
linear_acceleration_z -57.58281    14.40415   -3.998 7.40e-05 ***
orientation_x  -389.68966    174.10035   -2.238 0.025656 *
orientation_y   279.55776    125.60720    2.226 0.026498 *
orientation_z   23.13748     7.26534    3.185 0.001543 **
wind_angle     0.04790     0.05478    0.874 0.382320
flight_phase2  416.52073    21.22223   19.627 < 2e-16 ***
flight_phase3  -8.90241     33.48109   -0.266 0.790434
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 93.13 on 483 degrees of freedom
Multiple R-squared:  0.8516,    Adjusted R-squared:  0.8479
F-statistic: 231 on 12 and 483 DF,  p-value: < 2.2e-16
```

A stepwise regression was used to add and removed variables until it found an optimal model. It automatically excluded any statistically insignificant predictors from the model.

Stepwise (Final Model)

```
Call:
lm(formula = total_power ~ payload + velocity_z + linear_acceleration_z
+ orientation_x + orientation_y + orientation_z + flight_phase,
    data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-432.51  -23.87   -6.73    11.88   648.48

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -559.74857   139.69487  -4.007 7.11e-05 ***
payload         0.06542    0.01968   3.324 0.000954 ***
velocity_z     32.31301    9.04530   3.572 0.000389 ***
linear_acceleration_z -57.77913   14.21336  -4.065 5.60e-05 ***
orientation_x  -383.07474   172.54749  -2.220 0.026871 *
orientation_y   284.53190   121.71405   2.338 0.019806 *
orientation_z   23.82536    7.11858   3.347 0.000880 ***
flight_phase2   420.93503   13.59464  30.963 < 2e-16 ***
flight_phase3   -1.49111   10.27486  -0.145 0.884674
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92.84 on 487 degrees of freedom
Multiple R-squared:  0.8513,    Adjusted R-squared:  0.8489
F-statistic: 348.5 on 8 and 487 DF,  p-value: < 2.2e-16
```

The ANOVA test was used to compare the full model to the final model. ANOVA was used to drop variables during stepwise selection. The results seem to conclude that the full model was better.

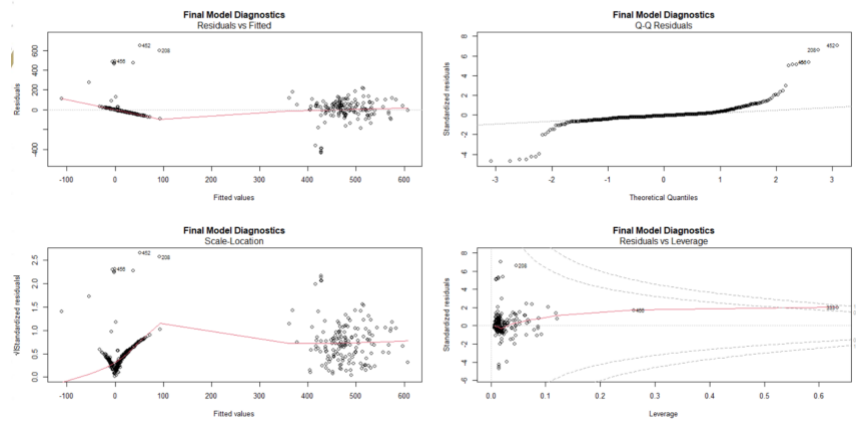
Stepwise (ANOVA)

```
--- ANOVA Model Comparison (Final vs Full) ---
> print(anova(final_model, full_model))
Analysis of Variance Table

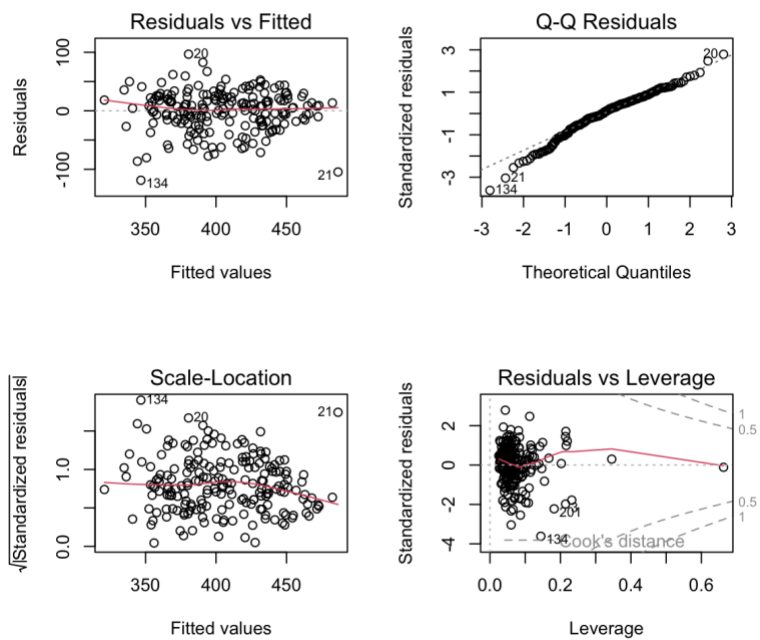
Model 1: total_power ~ payload + velocity_z + linear_acceleration_z +
orientation_x + orientation_y + orientation_z + flight_phase
Model 2: total_power ~ time + altitude + speed + payload + velocity_z +
linear_acceleration_z + orientation_x + orientation_y + orientation_
z +
wind_angle + flight_phase
    Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     487 4197479
2     483 4189083   4     8395.6 0.242 0.9145
```

The visual diagnostics for the residuals confirmed that the model is unstable. It should not be used to predict power consumption and is purely meant for explanatory purposes

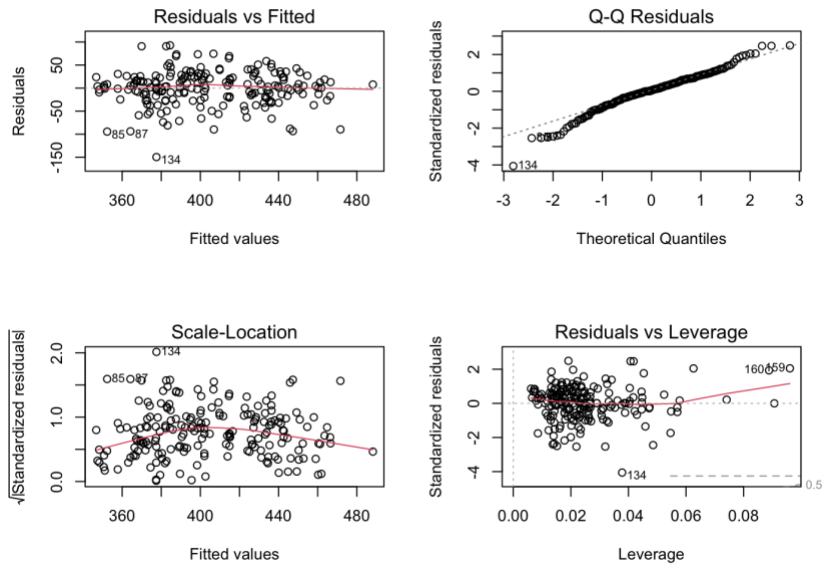
Stepwise (Diagnostics Plot)



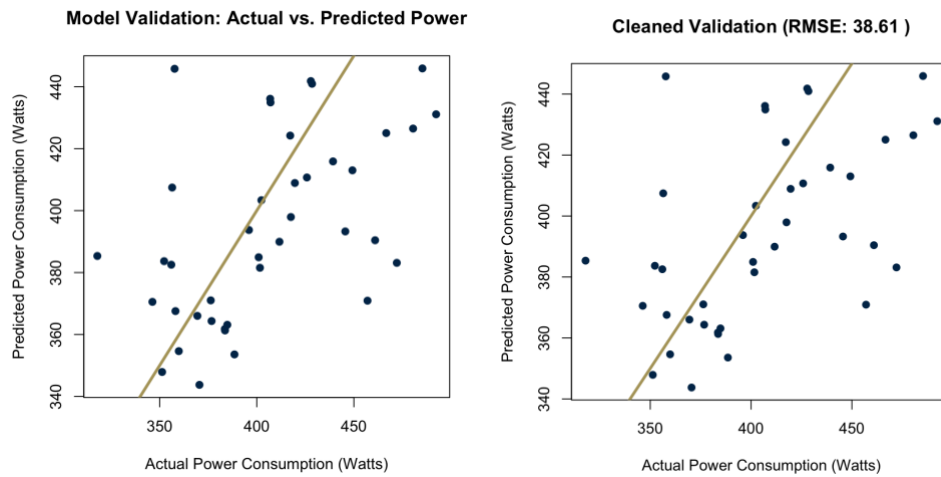
LASSO (Diagnostics Plot – Full Model)



LASSO Diagnostics Plot (Final Model – Reduced Model)



LASSO (RMSE)



LASSO (Coefficient Magnitude)

